

Ticket Booking with Consistency Check

Objective: Use a transaction plus a consistency condition to prevent over-booking.

CREATE TABLE events (

event_id **INT PRIMARY KEY**,

name **VARCHAR**(100),

total_seats **INT**,

booked_seats **INT**

);

INSERT INTO events **VALUES**

(1, 'Tech Conference', 100, 95);

Task

1. Write a transaction that tries to **book 10 seats** for event 1.
2. It should:
 - Read current booked_seats.
 - Check if booked_seats + 10 <= total_seats.
 - If yes, update and commit.
 - If no, rollback.
3. Assume your DB does not have procedural IF; use comments to describe the check logic if needed.

START TRANSACTION;

SELECT total_seats, booked_seats

FROM events

WHERE event_id = 1;

-- Suppose result: total_seats = 100, booked_seats = 95

-- Check: $95 + 10 \leq 100$? (FALSE)

-- Since condition fails, we should not update:

-- UPDATE events

-- SET booked_seats = booked_seats + 10

-- WHERE event_id = 1;

ROLLBACK;

SELECT * FROM events WHERE event_id = 1; -- still 95 booked

Possible successful variant (booking 3 seats):

sql

START TRANSACTION;

SELECT total_seats, booked_seats

FROM events

WHERE event_id = 1;

-- 95 + 3 <= 100 is TRUE

UPDATE events

SET booked_seats = booked_seats + 3

WHERE event_id = 1;

COMMIT;

SELECT * FROM events WHERE event_id = 1; -- now 98 booked

Explain how this transaction preserves the **consistency** of the events table